

RESPONSES TO TERM III PHYSICS 2025

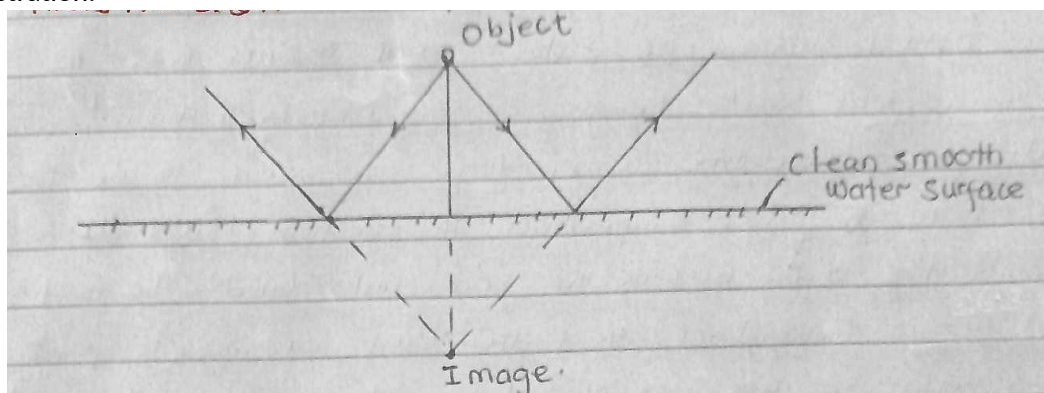
ITEM 1

- (a) The herd's man noticed himself due to **reflection** of light rays from plane or smooth water surface that **formed its virtual image in water**.

When he touch in water, the surface behave as a **rough or irregular surface** and therefore **irregular or diffuse reflection occurred**.

The reflected rays **are scattered in different directions** and at different angles hence no clear image is formed.

Illustration:

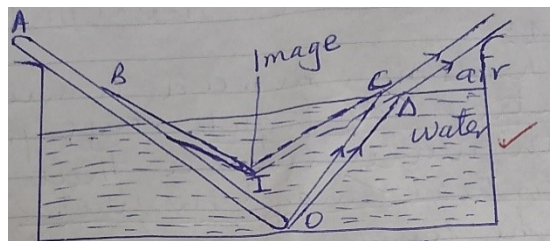


His stick appeared broken due to effects of **refraction** of light.

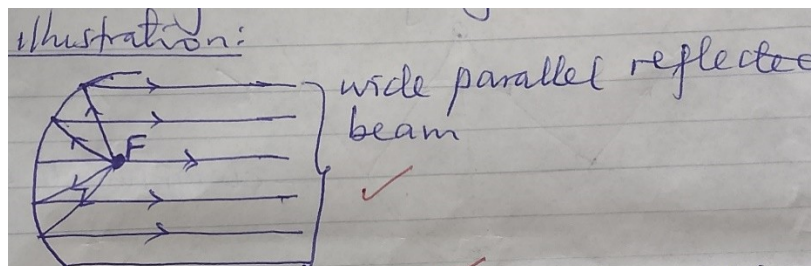
This is because as light travels from water to air, it **changes speed** and **bend** at the **air- water interface**. The light rays that allow you to see the submerged port of the stick travel from water into air.

Light will therefore **travel slower in water than in air** because **water is optically denser**.

When light passes from one medium to another, it **changes it's direction of travel** and this is refraction.



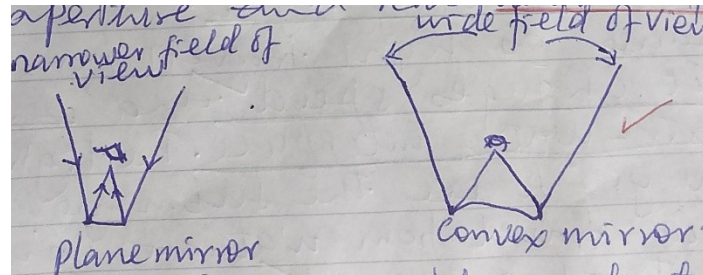
For boda boda rider, in the lighting system, a **concave mirror** is most appropriate because the purpose is **to produce an intense wide parallel beam** when the filament is placed at it's focus. In this way, it focuses light to a longer distance a head.



For riding, a **convex mirror** is most appropriate compared to plane mirror of the same aperture and has a **wide field of view**.

Narrow field of view

wide field of view



It also forms upright or erect image in all positions of the object.

From fig 1(a) $v = f\lambda$

$$2\lambda = 16$$

$$\lambda = \frac{16}{2} = 8\text{cm}$$

$$\lambda = \frac{8}{100} = 0.08\text{M}$$

$$F = \frac{330}{0.08} = 4125\text{HZ}$$

From fig 1(b), $f = \frac{1}{T}$

$$\frac{2T}{2} = \frac{0.00055}{2}, T = 0.000275$$

$$: f = \frac{1}{0.000275}$$

$$= 3636.364\text{HZ}$$

In both, sound signals, the frequencies are within the audible range and therefore the sound does not pose risks to the ears.

The difference in frequencies implies that the sound signals from the source was interfered with. This therefore means the unclear sound was due to interference, background noise.

The reverberation and echoes also interrupt the original sound and make it unclear and confused, disorganized.

(e) Applications of concepts like

- Reflection:
- ✓ In mirrors for grooming and security. ✓
- ✓ Vehicle mirrors for improved visibility. ✓

ANY TWO

- ✓ Periscopes for observing a round obstacles.
- ✓ Telescopes and microscopes for scientific observation.
- ✓ Solar lookers to concentrate sunlight for heat.
- ✓ Retro reflective materials for safety on roads and clothing's
- Refraction
 - ✓ In optical instruments like cameras, spectacles, telescope which use lenses to manipulate light for imaging or magnification.
 - ✓ Rainbow and twinkling of stars also use light bending as it passes different media.
 - ✓ In human eye to enable us focus light onto the retina for sight, an in optical fiber optics and prisms
 - ✓ Ie
 - ✓ In lenses and magnifying glasses in cameras microscopes telescopes to form images zoom in objects.
 - ✓ Spectacles and contact lenses to focus light properly for people with vision defects; improving their sight.
 - ✓ Projectors to creat and display enlarged images on a screen.
 - ✓ Apparent depth of water, objects submerged in water appear shallower than their actual depth as it moves from the water into air.
 - ✓ Applications of echoes.
 - ✓ In sonar, used in submarines and ships to detect objects under water by analysis reflected sound waves.
 - ✓ Ultra sound to create images of internal body structures.
 - ✓ Echolocation in bats and dolphins for hunting navigation.
 - ✓ In architectural design to control sound in spaces e.g concert halls.
 - ✓ Distance measurement by engineers to measure depth of wells, width of valleys or detect flaws like cracks by analysis the time it takes for sound to return.
 - ❖ Music and sound design , echoes and reverberation are used to create a sense of space and special effects in recorded music and sound.
 - ❖ Noise reduction: sound absorbing materials can be used to reduce unwanted echoes in vehicles or buildings improving charity and comfort.

Item 2

- (a) Due to **high ocean**✓**tides**.

This is caused by **gravitational**✓**pull of the moon** and sun on earth's oceans, creating **tidal bulges**✓**of water** on the sides of earth facing and opposite the moon.

- o As the earth rotates✓,different areas moves into these bulges resulting in light tides, therefore risky.
 - o A combination of these gravitational forces, along with earth's rotation and the effects of it's coastal lines and ocean floor determine the specific height and frequency of tides✓ daily.
- (b) Location is done by use of GPS (Global positioning satellites)
- A GPS powered device is **turned**✓**on**, for example smartphone, computer, GPS tracker.
 - The **device communicates**✓**with at least three satellite** and **GPS receiver like** that in the car tracking **picks**✓**signals** from these satellite which **send electromagnetic**✓**signals**, since radio waves travel at a speed of light hence **the receiver can compute**✓**the distances** of satellites from device and time ie (from satellite to receiver).
 - The system provides location in latitude, longitude and altitude. ✓

- Therefore, by knowing the distance from 4 or more satellites, the receiver determines its precise location on earth in **three dimensions**✓ **along latitude**, longitude and altitude.
- Location is determined by special geometry called **trilateration**✓ to locate actual position of people or device.
- The location is **displayed**✓ **on your screen** which is sent through the phone.

Also allow:

- **Monitoring satellites**, used **for surveillance** basically a self contained communication system with the ability **to receive signals from the earth** and **retransmit** those signals back with the **help of transponder**, an integrated receiver and **transmitter of radio signals**.
- Monitoring satellites use various instruments **to collect data from the environment**. They **capture images on earth** and **record videos**.

- The collected data is transmitted to the ground station where **it's processed** and **analysed** to provide information and predictions of where the **people are located**.

(c) Application of ocean tides:

- Generation of electricity using tidal energy for lighting✓.
- Distribution of minerals on water like sand for building purpose. ✓ **any two**
- Uniform temperature distribution in water. ✓
- Uniform distribution of planktons aquatic food for fish.
- Damarhkate coastal lines.
- Nutrient and oxygen circulation in coastal areas supporting aquatic life.
- Magnetic and electric effects to study the earth's interior.

Applications of GPS.

- In tracking system to locate lost items.
- In fleet management to locate exact position of different vehicles.
- Navigation to locate routes.

Item 3

- (a) A **radio active isotope** that emits **gamma**✓ **rays** such as technetium -99m is used to detect underground water pipe leakage because **gamma rays have a high penetrating power** ✓ **allowing them to travel through soil**✓ and **pipe** to be detected by **surface instrument like a GMT-tube**✓.
- ✓ **Gamma emitting radio isotope tracer is added**✓ **to the water** in the pipe. **The contaminated water seeps**✓ **out** at the leak, **concentrating**✓ **the radioactive** materials in the soil above the pipe.
 - ✓ A Geiger muller tube connected to aratemeter (counter) ✓ **is then moved**✓ **along the line of the pipe** ie above the pipe to and for several. Times to **scan the ground**✓, **detecting the increased gamma radiation** at the leak point and indicating the source of the leak.
 - ✓ So the highest count rate on the detector will occur when the Geiger-muller tube is positioned directly above✓ the leak, indicating the exact location of the underground leak.

The radioisotope used is supposed to have a short half life **to minimize**✓ **long term environmental risks**.

Gamma rays are also used as tracer due to lack of absorption gamma rays are less ✓ absorbed by materials **ensuring that the signal ✓ reaches the detector** above the ground.

The sample undergoes the decays:

$$200 \xrightarrow{t \frac{1}{2}} 100 \xrightarrow{t \frac{1}{2}} 50$$

$$2t \frac{1}{2} = 2 \times 6 = 12 \text{ hours}$$

Or

Activity	Time
200	0
100	6
50	12

The locals will wait for twelve hours before they use water from the underground pipe ✓.

Dangers associated with use of radio isotope.

- Leads to burning of skin and body tissues. ✓ any two
- Leads to damage deep in the body tissues and particularly body nuclei. ✓

Precautions / handling radio active material:

- Using forceps and kept in lead boxes to avoid contact with them. ✓
- Using protective gears like gloves to avoid contact. ✓ any two
- Working surfaces and storage areas such as tables tops and floors should be covered with absorbent materials to contain radioactive materials.

Applications of radio isotopes.

- Thickness gauging to monitor ✓ thickness of materials during manufacture. Any two
- Carbon dating in manufacture ✓ of carbohydrates.
- Uptake of fertilizer by plants.
- In space exploration used in nuclear batteries to generate electricity for space craft.

Item 4

(a)(i) Heat and noise are due to **friction**; ✓

Minimized by

- Lubrication of moving parts ✓
- Use of ball bearings ✓ or rollers at axles. ✓
- Making surface smooth

(ii) Maximum load carried by convey or,

$$\text{From efficiency} = \frac{\text{work out put}}{\text{work in put}} \times 100$$

$$\text{Eff} = \frac{I \times I_d}{E \times E_d} \times 100\%$$

$$50 = \frac{LXW}{150 \times 14} \times 100 \quad \therefore L = 75\text{N} \checkmark$$

Assumption made: load distance is equal to the distance moved by effort $\checkmark$ (driving force)

(b) System got a fault and stopped working. $\checkmark$

Might be leakage from pipe

- $\checkmark$ Evaporation, therefore more $\checkmark$ water should be added into the drum. Any three
- $\checkmark$ Heating source might have gone off or might be produced less heat, therefore should be checked. $\checkmark$
- $\checkmark$ Rust and a layer of dust particles suspected on metallic heater should be removed.
- $\checkmark$ Source pan made of good conductor to minimize rate of evaporation.

(c) Applications friction as physics concept;

- $\checkmark$ Sharpening $\checkmark$ knife for peeling. $\checkmark$
- $\checkmark$ Walking $\checkmark$ to reach destination. $\checkmark$ any three
- $\checkmark$ Lighting $\checkmark$ match box to produce $\checkmark$ fire/light
- $\checkmark$ Lubrication to minimize friction is applied in machinery to minimize wear and tear of engine parts.

ITEM 5

- (a) It spread on wider piece of cloth due to **weak adhesive $\checkmark$ forces** compared to **strong cohesive forces** of the fabrics threads causing the liquid to move freely and penetrate the fabrics structure.
- $\checkmark$ Paraffin also has a **low viscosity $\checkmark$** allowing it to **spread quickly and easily into the small spaces $\checkmark$ between** clothing fibers, rather than staying in a compact drop. I.e cohesive forces within paraffin itself are stronger than adhesive forces holding it to the cloth.
 - $\checkmark$ **Capillary attraction $\checkmark$** in fabric, cloth with its woven threads has tiny spaces called capillaries. Because paraffin is a low viscosity liquid with weak adhesion to the fabric these **capillaries readily draw $\checkmark$** the paraffin into the fabric.
 - $\checkmark$ The stain also got off quickly when done as friend advised because, **detergent lower $\checkmark$ surface tension $\checkmark$** of water hence removing dirt easily.
 - $\checkmark$ Warm water **weakens $\checkmark$ the inter molecular forces of attraction** between the stained particle and cloth hence removing the stain easily.
- (b) Establishing correct temperature of water to wash the stain. $\checkmark$

Assume the ice point at 0cm mark, and steam point at 28cm, then

$$\Theta = \frac{x}{y} \times 100 \checkmark$$

$$= \frac{14}{28} \times 100 \checkmark : \Theta = 50 \checkmark ^\circ\text{C} \checkmark$$

Conclusion:

The temperature appropriate to remove the stain was between 46-51 $^\circ\text{C}$ $\checkmark$ and the calculated value is 50 $^\circ\text{C}$, therefore the stain was removed. $\checkmark$

(c) Applications of physics concepts:

Any three

(i) Capillarity;

- Absorption of liquids ✓ by paper, towels sponges to clear ourselves, allowing fluids.
- To be transferred from surface to the towel.
- In lamp wicks and candles, allowing the flame to burn continuously ✓ as paraffin rises in wicks ie transfer liquid from store to flame.
- In plants and trees to draw water from soil through their roots ✓ and transport it to their stems, leaves to provide essential nutrients for growth.
- Tear drainage from eyes preventing excess moisture build up.
- Ink travel in fountain pen reservoir to nib enabling writing.
- In towels to remove water from body.

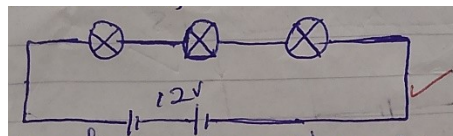
(ii) Surface tension.

- Floating objects like needle to float on water. ✓ **any two**
- Water resistant fabrics (tent effect) to bridge ✓ the tiny pores in fabrics to prevent water pass through.
- Formation of soap bubble to pull the liquid film to a minimal surface area.
- Cleaning.
- Lung function during breathing.

ITEM 6

(a) Bulbs can either be connected in series or parallel.

In series,



Effective resistance $R=2+2+2$ ✓

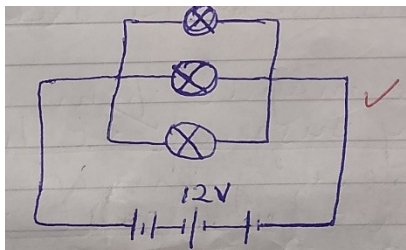
$$=6\Omega \text{ ✓}$$

From $V=IR$

$$12 = I \times 6 \text{ ✓}$$

$$I = \frac{12}{6} = 2A$$

In parallel



Effective resistance R is given by

$$\frac{I}{R} = \frac{I}{R_1} + \frac{I}{R_2} + \frac{I}{R_3} = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}$$

$$\therefore R = \frac{2}{3} \Omega$$

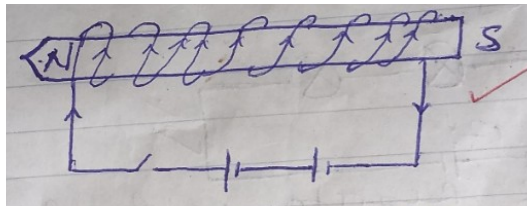
From $V=IR$

$$R = I \times \frac{2}{3}, I = 18A$$

In parallel connection **more current** is drawn from the battery hence shorter battery life span, therefore **brighter** light.

In series connection, **lower current is drawn from the battery** hence longer life span, therefore **dimmer** light.

(b) The



students meet need to **make a magnet** using electrical method.

- Wind the insulated copper wire on a steel nail to form a solenoid and connected it to the ends of the accumulator. ✓
- When the switch is closed ✓, **current flows through the solenoid creating a magnetic field around the coil.**
- **This magnetizes the steel nail** in the solenoid.
- Remove the magnet flow the solenoid and **freely suspend** it horizontally using a piece of thread. ✓
- The magnet **will rest in the north** south direction. ✓
- The students will then **move following the south seeking** pole of the permanent magnet made to go back to school. ✓

(c) **Applications of magnets.**

Any two

- In refrigerators to keep refrigerator doors tightly closed and hold papers and small items on the doors. ✓
- In electronics like computers and speakers headphones, microphones to convert signals into sound and vice-versa. ✓
- In computers and phones to store data and in mobile devices for various components.
- In compasses, to show direction magnetic strips on credit cards for storing data.
- In recycling to separate ferrous metals from other materials.
- In food processing to remove metallic components/ contaminants from grains and other food products.
- In manufacturing to lift heavy objects and convey them in assembly lines.
- In medical devices like MRI machines that use strong magnetic fields to generate detailed images of human body.
- Safety in industrial setting to separate iron particles to remove metallic objects from animals.

- In electric motor and generators to convert electrical energy into mechanical motion.

ITEM 7

(a)	Magnetic scrapper system. ❖ Allow lifting magnet which operates based on the principle of electromagnetism. ❖ Explanation ❖ When an electric current passes through a coil of wire, it generates a magnetic field. ❖ The magnetic field can attract Ferro-magnetic materials but will not affect non-magnetic materials. ❖ The magnetic scrapper uses the magnetic field generated by the current to attract and separate magnetic metals from non-magnetic ones. ❖ As the magnetic scrapper moves over a mixture of materials the magnetic metal are drawn towards the magnet and separate them from the rest.	ID	02
(b)	Energy consumed <i>per day</i> = <i>power</i> $\times$ <i>time</i> $\downarrow \frac{100}{1000} \times 6$ $\downarrow 0.6 \text{ kwh}$ Energy reduction $\downarrow \frac{25}{100} 0.6$ $\downarrow 0.15 \text{ kwh}$ The new energy consumed = $0.6 - 0.15$ $= 0.45 \text{ kwh}$	EX	07
(c)	To avoid over heating therefore the power consumed should be: New power = new energy consumed $\times$ time. $= 0.45 \times 6$ $= 0.75 \text{ kwh. (allow 75w)}$ To avoid overheating the system should be ideally consume 75w of power. To enhance the system; • Use more efficient components e.g more efficient, energy supply and coil with lower resistance/ coil with many windings. (any two)	MC	09

	• Increasing number of turns in the coil will increase the magnetic field strength. • Increasing the current flowing through the coil can also increase the magnetic field strength. • Using soft iron core to make a coil. OTHER APPLICATIONS. ✓ Electric bells. ✓ Magnetic relays ✓ Moving coil loud speaker ✓ Telephone receivers	AP TOTAL	O2 20kps
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(Any two)